



Midterm-1 Exam
Electromagnetic Theory
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Duration: 75 min

Q1 [50 points]: A fixed-diameter cylindrically symmetric (volumetric) charge configuration extends to infinity in the $\pm z$ directions; the radius of the cylinder is $\rho = a$. Inside the radius b , where $b < a$, the charge density has a parabolic dependence on radius while over the range $b < \rho < a$ outside that radius, the volumetric charge density is zero.

$$\rho_v = \begin{cases} \rho_o \frac{\rho^2}{b^2} & \rho < b \\ 0 & b < \rho < a \end{cases}$$

(a) [40 points] Use the axial symmetry and Gauss' integral law to show that $\mathbf{E}$ in the two regions is

$$\mathbf{E} = \begin{cases} \frac{\rho_o \rho^3}{4\epsilon_o b^2} \mathbf{a}_\rho & \rho < b \\ \frac{\rho_o b^2}{4\epsilon_o \rho} \mathbf{a}_\rho & b < \rho < a \end{cases}$$

(b) [10 points] Outside the cylinder at $\rho > a$, the electric field is zero, $\mathbf{E} = 0$. This is made possible by implanting deliberately a surface charge density on the outer surface of the cylinder. Using the boundary condition, $\epsilon_1 \mathbf{E}_1 - \epsilon_2 \mathbf{E}_2 = \rho_s \mathbf{a}_n$, obtain the surface charge density at $\rho = a$.

Note: Since, no dielectric material is specified here, you may assume that $\epsilon_1 = \epsilon_2 = \epsilon_0$, where ϵ_0 is the permittivity of the free space.

Q2 [20 points]: A cylindrical slab has a polarization given by $\mathbf{P} = \rho_0 \sqrt{\rho} \mathbf{a}_\rho$. Find the polarization volumetric charge density ρ_{pv} inside the slab and its polarization surface charge density ρ_{ps} . Note: $\rho_{ps} = \mathbf{P} \cdot \mathbf{a}_n$ and $\rho_{pv} = -\nabla \cdot \mathbf{P}$.

Q3 [30 points]: Point charges $Q_1 = 2\text{nC}$, $Q_2 = -3\text{nC}$, $Q_3 = 4\text{nC}$, and $Q_4 = -2\text{nC}$ are positioned one at a time and in that order at $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, -1)$, and $(0, 0, 1)$, respectively. Calculate the energy in the system after each charge is positioned.

Q4 [10 points]: Inside the cylinder of rectangular cross-section, the potential is given as

$$V = \frac{\rho_o}{\epsilon_o} \cos \frac{\pi}{a} x \cos \frac{\pi}{b} y$$

Find $\mathbf{E}$.